

ANSHUMAN ACHARYA

Hyderabad, Telangana, India

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EDUCATION

- **Gandhi Institute for Technological Advancements** Bhubaneswar, Odisha
Bachelors of Technology in Computer Science & Information Technology; CGPA: 7.73 2021 – 2025

SKILLS

Programming: Python, SQL, Java, JavaScript, C

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Model Evaluation, Feature Engineering

Deep Learning & Computer Vision: TensorFlow, Keras, CNN, RNN, LSTM, OpenCV, YOLOv8, U-Net, Image Segmentation, Object Detection

AI / NLP / Automation: LangChain, FAISS, RAG, Prompt Engineering, Gemini API, n8n

Databases: MySQL, PostgreSQL, MongoDB, DBMS

Tools / Platforms: Git, GitHub, Jupyter Notebook, VS Code, NetBeans, Spyder, Label Studio, Roboflow

EXPERIENCE

Han Digital Jul 2026 – Sep 2026
Freelance Process Associate (AI Data Annotation) Hyderabad, India

- Annotate data for AI projects according to project-specific guidelines and quality standards.
- Maintain **95%+ annotation quality** based on regular audits and feedback.

PROJECTS

- **AI-Based Detection of Missing and Damaged Road Markings for Road Safety**

Built an end-to-end AI-powered system for automated road damage and lane marking detection using deep learning-based object detection.

Workflow: Multi-source dataset collection, COCO-to-YOLO annotation conversion, preprocessing and augmentation, training separate YOLOv8 models for road damages and lane markings, combined inference development, model evaluation using Precision, Recall, and mAP metrics, and deployment of an interactive Streamlit dashboard with analytics and prediction monitoring.

Tech Stack: Python, PyTorch, YOLOv8 (Ultralytics), OpenCV, Streamlit, Plotly, Roboflow, NumPy, Pandas, Google Colab.

- **AI-Based Road Crack Detection & Severity Assessment System**

Built an end-to-end deep learning-based system for automated road crack detection and severity assessment using image segmentation techniques.

Workflow: Multi-source dataset collection (12K+ images from Kaggle, research papers, GitHub), annotation with Label Studio and Roboflow, preprocessing and augmentation, segmentation using YOLOv8 and U-Net, model comparison using mAP, IoU, Precision and Recall, and deployment strategy design.

Tech Stack: Python, PyTorch, YOLOv8 (Ultralytics), U-Net, OpenCV, Label Studio, Roboflow, NumPy, Pandas.

- **Retrieval-Augmented Generation (RAG) Knowledge-Based Chatbot**

Built an end-to-end Retrieval-Augmented Generation (RAG) chatbot that retrieves semantically relevant document chunks and generates grounded responses using a large language model.

Workflow: Knowledge base ingestion, document chunking, embedding generation, FAISS vector indexing, semantic similarity retrieval, context-constrained prompt engineering, Gemini-based response generation.

Tech Stack: Python, LangChain (LCEL), FAISS, HuggingFace (MiniLM), Google Gemini API.

CERTIFICATIONS

- **Certification in Data Science & AI** 360DigiTMG
by 360DigiTMG (June 2025 – Present)
- **Databases and SQL for Data Science with Python** Coursera
by IBM Jan 2025
- **Fundamentals of Generative AI** Coursera
by Amazon Web Services July 2024